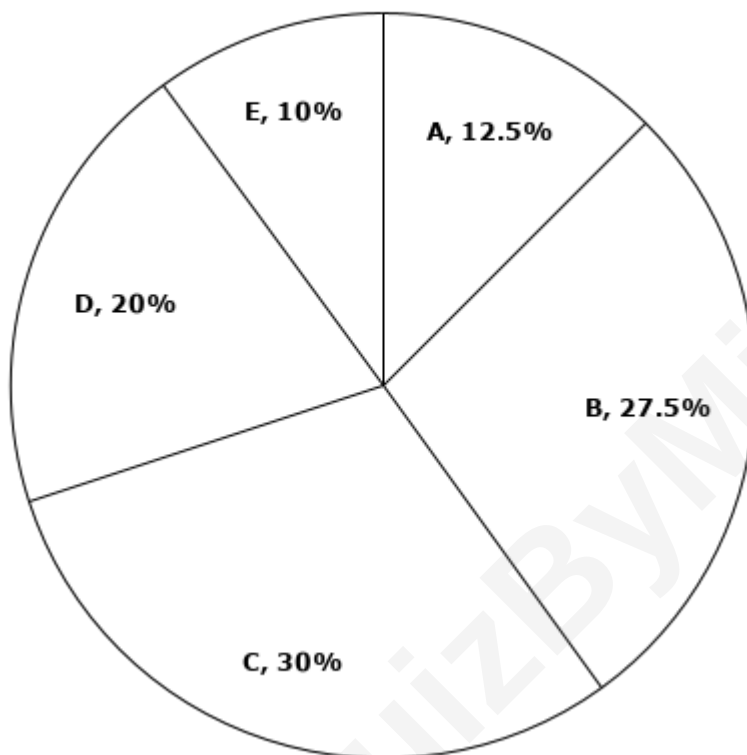


1. Questions

Study the following information carefully and answer the questions given below.

The given pie chart shows the percentage distribution of the total number of units shipped by five different brands : A, B, C, D, and E in a fiscal year. The total number of units shipped by all five brands together is 4000.

Distribution of the total number of units shipped by five different brands



Find the sum of 45% of the total number of units shipped by Brand B and the total number of units shipped by Brand E.

- a. 885
- b. 905
- c. 795
- d. 895
- e. 995

2. Questions

What is the ratio of twice the total number of units shipped by Brand A to 1/3rd of the total number of units shipped by Brand C?

- a. 5:3
- b. 2:5

- c. 3:2
- d. 5:4
- e. 5:2

3. Questions

Find the difference between the total number of units shipped by Brand D and 60% of $\frac{1}{4}$ th of the total number of units shipped by Brand C.

- a. 520
- b. 620
- c. 720
- d. 680
- e. 640

4. Questions

Brand B ships its total number of units using 3 Type-X and 2 Type-Y containers, and Brand C uses 1 Type-X and 4 Type-Y containers. Given that Type-X and Type-Y containers have different unit capacities, find the total number of units in Type-X containers used by brands B and C together.

- a. 1200
- b. 1500
- c. 800
- d. 1000
- e. 1250

5. Questions

$\frac{3}{4}$ th of the total number of units shipped by Brand C is what percent less than 2.5 times the total number of units shipped by Brand E?

- a. 10%
- b. 12.5%
- c. 20%
- d. 11.11%
- e. 15%

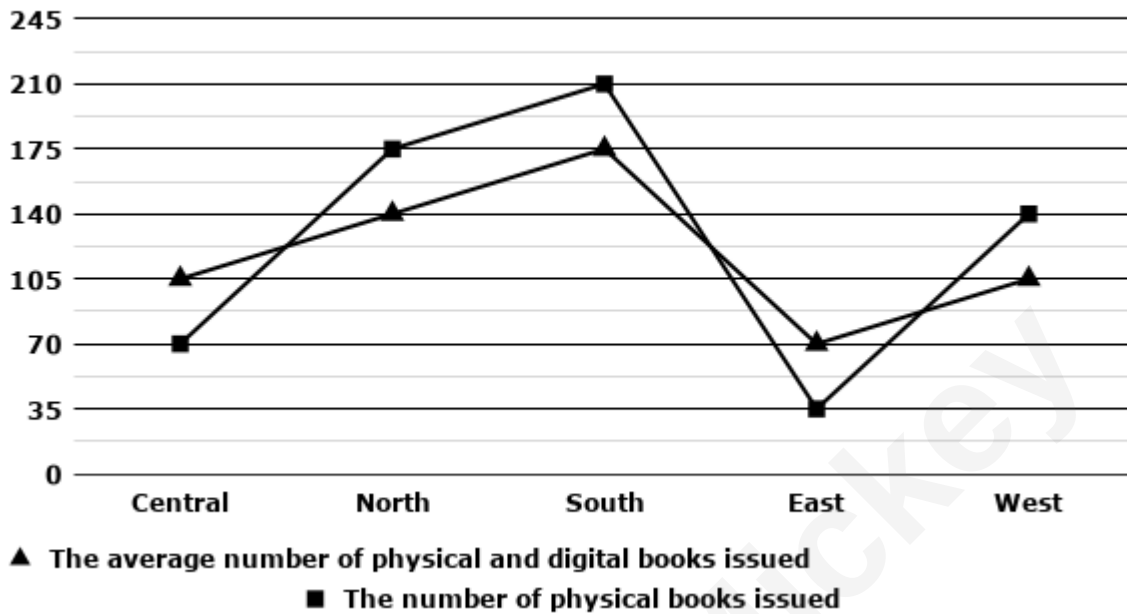
6. Questions

Study the following information carefully and answer the questions given below.

The given line graph shows the average number of physical books and digital books issued weekly by five

libraries : Central, North, South, East, and West along with the total number of physical books issued by each library.

Distribution of physical books and digital books issued weekly by five libraries



Note: The total number of books issued = The number of physical books issued + The number of digital books issued.

Find the average of 80% of the number of physical books issued by the South library, the number of digital books issued by the North library, and 1.5 times the number of digital books issued by the West library.

- 130
- 116
- 120
- 126
- 136

7. Questions

Three times of $\frac{2}{7}$ th of the number of physical books issued by the Central library is what percentage of the total number of books issued by the East library?

- 50%
- 57.14%
- 28.56%
- 35%
- 42.85%

8. Questions

The South library found that 'x' digital books remained unreturned. If 20% of the total number of books issued by the South library were returned in a physical format, and the sum of the unreturned digital books and the returned physical books is 95, find the value of x.

- a. 25
- b. 45
- c. 30
- d. 20
- e. 35

9. Questions

The total fine collected by the East library for all its issued physical books and all its issued digital books is Rs. 1050. The total fine collected by the West library for all its issued physical books and all its issued digital books is Rs. 2450. Find the total late renewal fine for 10 physical books.

- a. Rs. 160
- b. Rs. 200
- c. Rs. 150
- d. Rs. 100
- e. Rs. 50

10. Questions

40% of the total number of physical books issued by the North Library is what fraction of the total number of digital books issued by the East Library after it is increased by one-third of itself?

- a. $\frac{2}{3}$
- b. $\frac{1}{2}$
- c. $\frac{3}{4}$
- d. $\frac{1}{3}$
- e. $\frac{1}{4}$

11. Questions

Study the following information carefully and answer the questions given below.

In a university of 600 students, 80 do not participate in any of the three available athletic programs: Cricket, Football, and Tennis. The number of students participating only in Football is 100, which is twice the number of those playing both Cricket and Football but not Tennis, and the 40 students participating in all three sports are 20 fewer than those participating in both Football and Tennis but not Cricket. The number of students participating in both Cricket and Tennis but not Football is equal to the number of

students participating in both Football and Tennis but not Cricket. The number of students participating only in Cricket is 40 more than the number of students participating only in Tennis.

What is the ratio of 80% of the number of students participating only in Cricket to the number of students participating in both Cricket and Tennis but not Football?

- a. 5:4
- b. 4:3
- c. 3:5
- d. 5:2
- e. 5:3

12. Questions

Find the product of $\frac{4}{5}$ th of the number of students participating only in Tennis and 1.2 times the number of students participating in both Cricket and Football but not Tennis.

- a. 4080
- b. 4880
- c. 4180
- d. 4480
- e. 4280

13. Questions

What is the difference between 2.5 times the number of students who do not participate in any of these three sports and 150% of the number of students participating in both Football and Tennis but not Cricket?

- a. 100
- b. 90
- c. 110
- d. 120
- e. 130

14. Questions

If the sum of the value of y and the number of students participating only in Football equals 3.5 times the number of students participating in both Cricket and Tennis but not Football, find the value of y .

- a. 90
- b. 110

- c. 100
- d. 130
- e. 120

15. Questions

A is 'x' years older than B, and B's age 'x' years ago was half of A's present age. C is 8 years older than twice the present age difference of A and B. If the sum of their present ages is 62 years, find the value of x.

- a. 8
- b. 5
- c. 6
- d. 4
- e. 7

16. Questions

A merchant sells two articles at the same price, gaining 20% profit on the first article. If he makes no profit or loss on the overall transaction, the cost price of the second article is Rs. 1200 more than the first. Find the selling price of each article.

- a. Rs. 2400
- b. Rs. 3200
- c. Rs. 3600
- d. Rs. 4800
- e. Rs. 2880

17. Questions

A vessel is filled with milk. 20% of the milk is replaced with water, and this process is repeated once more. If the final mixture contains 28 liters more milk than water, find the quantity of mixture replaced each time.

- a. 25 liters
- b. 20 liters
- c. 15 liters
- d. 30 liters
- e. 40 liters

18. Questions

P starts a business with Rs. p. Q joins after 't' months with Rs. 2p, and R joins after another 't' months with Rs. 3p. At the end of the year, P and R receive equal shares of the profit. If the total annual profit is Rs. 8000, find Q's share of the profit.

- a. Rs. 3600
- b. Rs. 2400
- c. Rs. 4000
- d. Rs. 3200
- e. Rs. 2800

19. Questions

A sum 'P' invested at 20% p.a. compound interest for 2 years yields Rs. 240 less interest than the sum '(P + 2000)' invested at 12% p.a. simple interest for 3 years. Find the value of 2P.

- a. 13500
- b. 12000
- c. 10000
- d. 12500
- e. 15000

20. Questions

A square park has a 2-meter wide path inside the borders. If the numerical value of the path's area is four times the numerical value of the park's inner perimeter, find the total area of the park.

- a. 64 m^2
- b. 100 m^2
- c. 36 m^2
- d. 49 m^2
- e. 81 m^2

21. Questions

What value will come in place of the question mark (?) in the following question?

$$(?^2 + 13 * 8) / \sqrt{196} + \sqrt{144} = 27 + 3^3$$

- a. 34
- b. 12

- c. 22
- d. 16
- e. 28

22. Questions

$$18 * 4 + 20\% \text{ of } 150 + 2(1/4) * ? = 15^2 - 15$$

- a. 62
- b. 48
- c. 54
- d. 42
- e. 38

23. Questions

$$? - 17^2 = (14 * 2 + 12) * (15/2)$$

- a. 609
- b. 569
- c. 599
- d. 589
- e. 579

24. Questions

$$?^3 - 5(1/3) * 27 + 11^2 = 20^2 + 89$$

- a. 8
- b. 16
- c. 4
- d. 6
- e. 12

25. Questions

$$480/\sqrt{?} + 11^2 + 8^2 = 14^2 + 29$$

- a. 134
- b. 124

- c. 154
- d. 164
- e. 144

26. Questions

What approximate value will come in place of the question mark (?) in the following question?

$$(?^2 - 22.95 * 4.02) / \sqrt{360.99} = 14.98 + 8.05$$

- a. 13
- b. 23
- c. 50
- d. 41
- e. 33

27. Questions

$$(419.98 - ?) / (11.02 / 3.99) = 15.95 * 5.03$$

- a. 170
- b. 225
- c. 200
- d. 185
- e. 210

28. Questions

$$?^2 + 13.98^2 = 39.95\% \text{ of } 650.05 + 15.97 * 5.02$$

- a. 48
- b. 35
- c. 60
- d. 12
- e. 22

29. Questions

$$(?^2 + 18.99) / (25.02 / 6.98) = 15.95 + 12.04$$

- a. 9
- b. 19

- c. 49
- d. 39
- e. 29

30. Questions

$$? * 11.95 - \sqrt{1023.98} = 34.98 * 3.99 + 20.02$$

- a. 26
- b. 44
- c. 9
- d. 35
- e. 16

31. Questions

What will come in the place of question mark (?) in the following number series?

10, 14, ?, 48, 97, 218

- a. 21
- b. 23
- c. 25
- d. 19
- e. 33

32. Questions

5, 11, 31, 127, ?, 3791

- a. 635
- b. 621
- c. 631
- d. 641
- e. 625

33. Questions

?, 60, 76, 98, 126

- a. 48
- b. 46

- c. 54
- d. 50
- e. 40

34. Questions

48, ?, 72, 36, 108, 54

- a. 24
- b. 28
- c. 26
- d. 22
- e. 34

35. Questions

100, 106, 130, ?, 310, 520

- a. 180
- b. 210
- c. 190
- d. 200
- e. 170

36. Questions

Find the wrong term in the given series.

15, 33, 54, 78, 107, 135

- a. 15
- b. 107
- c. 135
- d. 33
- e. 54

37. Questions

100, 112, 136, 172, 222, 280

- a. 280
- b. 172

- c. 112
- d. 136
- e. 222

38. Questions

8, 14, 24, 60, 180

- a. 8
- b. 60
- c. 14
- d. 180
- e. 24

39. Questions

10, 11, 38, 163, 500

- a. 500
- b. 11
- c. 38
- d. 163
- e. 10

40. Questions

25, 34, 46, 60, 76

- a. 46
- b. 76
- c. 60
- d. 25
- e. 34

41. Questions

In each of the following questions, two equations are given. You have to solve both the equations to find the relation between x and y.

I). $3x^2 - 38x + 115 = 0$

II). $2y^2 - 15y + 28 = 0$

- a. $x = y$ or the relationship cannot be established
- b. $x \leq y$
- c. $x > y$
- d. $x < y$
- e. $x \geq y$

42. Questions

I). $2x^2 + 31x + 119 = 0$

II). $3y^2 + 25y + 52 = 0$

- a. $x < y$
- b. $x > y$
- c. $x \leq y$
- d. $x \geq y$
- e. $x = y$ or the relationship cannot be established

43. Questions

I). $2x^2 - 23x + 65 = 0$

II). $2y^2 - 19y + 45 = 0$

- a. $x \leq y$
- b. $x = y$ or the relationship cannot be established
- c. $x < y$
- d. $x > y$
- e. $x \geq y$

44. Questions

I). $3x^2 - 7x - 40 = 0$

II). $y^2 - 14y + 45 = 0$

- a. $x \geq y$
- b. $x < y$
- c. $x \leq y$
- d. $x = y$ or the relationship cannot be established

e. $x > y$

45. Questions

I). $2x^2 - 27x + 88 = 0$

II). $y^2 - 13y + 42 = 0$

- a. $x > y$
- b. $x = y$ or the relationship cannot be established
- c. $x \geq y$
- d. $x \leq y$
- e. $x < y$

Explanations:

1. Questions

Brand A

The total number of units shipped = 12.5% of 4000 = $(12.5/100) * 4000 = 500$

Brand B

The total number of units shipped = 27.5% of 4000 = $(27.5/100) * 4000 = 1100$

Brand C

The total number of units shipped = 30% of 4000 = $(30/100) * 4000 = 1200$

Brand D

The total number of units shipped = 20% of 4000 = $(20/100) * 4000 = 800$

Brand E

The total number of units shipped = 10% of 4000 = $(10/100) * 4000 = 400$

Brands	The number of units shipped
A	500
B	1100
C	1200
D	800
E	400

Answer: D

The total number of units shipped by Brand B = 1100

45% of the total number of units shipped by Brand B = $(45/100) * 1100 = 495$

The total number of units shipped by Brand E = 400

The required sum = $495 + 400 = 895$

Hence, the correct answer is option D.

2. Questions

Brand A

The total number of units shipped = 12.5% of 4000 = $(12.5/100) * 4000 = 500$

Brand B

The total number of units shipped = 27.5% of 4000 = $(27.5/100) * 4000 = 1100$

Brand C

The total number of units shipped = 30% of 4000 = $(30/100) * 4000 = 1200$

Brand D

The total number of units shipped = 20% of 4000 = $(20/100) * 4000 = 800$

Brand E

The total number of units shipped = 10% of 4000 = $(10/100) * 4000 = 400$

Brands	The number of units shipped
A	500
B	1100
C	1200
D	800
E	400

Answer: E

The total number of units shipped by Brand A = 500

Twice the total number of units shipped by Brand A = $2 * 500 = 1000$

The total number of units shipped by Brand C = 1200

1/3rd of the total number of units shipped by Brand C = $(1/3) * 1200 = 400$

The required ratio = $1000:400 = 5:2$

Hence, the correct answer is option E.

3. Questions

Brand A

The total number of units shipped = 12.5% of 4000 = $(12.5/100) * 4000 = 500$

Brand B

The total number of units shipped = 27.5% of 4000 = $(27.5/100) * 4000 = 1100$

Brand C

The total number of units shipped = 30% of 4000 = $(30/100) * 4000 = 1200$

Brand D

The total number of units shipped = 20% of 4000 = $(20/100) * 4000 = 800$

Brand E

The total number of units shipped = 10% of 4000 = $(10/100) * 4000 = 400$

Brands	The number of units shipped
A	500
B	1100
C	1200
D	800
E	400

Answer: B

The total number of units shipped by Brand D = 800

The total number of units shipped by Brand C = 1200

The 1/4th of the total number of units shipped by Brand C = $(1/4) * 1200 = 300$

The 60% of this value = $(60/100) * 300 = 180$

The required difference = $800 - 180 = 620$

Hence, the correct answer is option B.

4. Questions

Brand A

The total number of units shipped = 12.5% of 4000 = $(12.5/100) * 4000 = 500$

Brand B

The total number of units shipped = 27.5% of 4000 = $(27.5/100) * 4000 = 1100$

Brand C

The total number of units shipped = 30% of 4000 = $(30/100) * 4000 = 1200$

Brand D

The total number of units shipped = 20% of 4000 = $(20/100) * 4000 = 800$

Brand E

The total number of units shipped = 10% of 4000 = $(10/100) * 4000 = 400$

Brands	The number of units shipped
A	500
B	1100
C	1200
D	800
E	400

Answer: C

The total number of units shipped by Brand B = 1100

The total number of units shipped by Brand C = 1200

Let the capacity of a Type-X container be 'x' and a Type-Y container be 'y'.

$$3x + 2y = 1100 \text{ --- (1)}$$

$$1x + 4y = 1200 \text{ --- (2)}$$

From equation (2)

$$x = 1200 - 4y$$

Substituting the value of x into the equation (1):

$$3 * (1200 - 4y) + 2y = 1100$$

$$3600 - 12y + 2y = 1100$$

$$2500 = 10y$$

$$y = 250$$

Substituting the value of y,

$$x = 1200 - 4 * 250$$

$$x = 1200 - 1000 = 200$$

The total number of units in Type-X containers used by brands B and C together = $3x + x = 4x = 4 * 200 = 800$

Hence, the correct answer is option C.

5. Questions

Brand A

The total number of units shipped = 12.5% of 4000 = $(12.5/100) * 4000 = 500$

Brand B

The total number of units shipped = 27.5% of 4000 = $(27.5/100) * 4000 = 1100$

Brand C

The total number of units shipped = 30% of 4000 = $(30/100) * 4000 = 1200$

Brand D

The total number of units shipped = 20% of 4000 = $(20/100) * 4000 = 800$

Brand E

The total number of units shipped = 10% of 4000 = $(10/100) * 4000 = 400$

Brands	The number of units shipped
A	500
B	1100
C	1200
D	800
E	400

Answer: A

The total number of units shipped by Brand C = 1200

3/4th of the total number of units shipped by Brand C = $(3/4) * 1200 = 900$

The total number of units shipped by Brand E = 400

2.5 times the total number of units shipped by Brand E = $2.5 * 400 = 1000$

The difference = $1000 - 900 = 100$

The required percentage = $(100/1000) * 100 = 10\%$

Hence, the correct answer is option A.

6. Questions

The total number of books issued = The average number of physical and digital books issued * 2

The number of digital books issued = The total number of books issued - The number of physical books issued

Central

The total number of books issued = $105 * 2 = 210$

The number of physical books issued = 70

The number of digital books issued = $210 - 70 = 140$

North

The total number of books issued = $140 * 2 = 280$

The number of physical books issued = 175

The number of digital books issued = $280 - 175 = 105$

South

The total number of books issued = $175 * 2 = 350$

The number of physical books issued = 210

The number of digital books issued = $350 - 210 = 140$

East

The total number of books issued = $70 * 2 = 140$

The number of physical books issued = 35

The number of digital books issued = $140 - 35 = 105$

West

The total number of books issued = $105 * 2 = 210$

The number of physical books issued = 140

The number of digital books issued = $210 - 140 = 70$

Libraries	The total number of books issued	The number of physical books issued	The number of digital books issued
Central	210	70	140
North	280	175	105
South	350	210	140
East	140	35	105
West	210	140	70

Answer: D

The number of physical books issued by the South library = 210

80% of this value = $(80/100) * 210 = 168$

The number of digital books issued by the North library = 105

The number of digital books issued by the West library = 70

1.5 times of this value = $1.5 * 70 = 105$

The sum of the values = $168 + 105 + 105 = 378$

The required average = $378/3 = 126$

Hence, the correct answer is option D.

7. Questions

The total number of books issued = The average number of physical and digital books issued * 2

The number of digital books issued = The total number of books issued - The number of physical books issued

Central

The total number of books issued = $105 * 2 = 210$

The number of physical books issued = 70

The number of digital books issued = $210 - 70 = 140$

North

The total number of books issued = $140 * 2 = 280$

The number of physical books issued = 175

The number of digital books issued = $280 - 175 = 105$

South

The total number of books issued = $175 * 2 = 350$

The number of physical books issued = 210

The number of digital books issued = $350 - 210 = 140$

East

The total number of books issued = $70 * 2 = 140$

The number of physical books issued = 35

The number of digital books issued = $140 - 35 = 105$

West

The total number of books issued = $105 * 2 = 210$

The number of physical books issued = 140

The number of digital books issued = $210 - 140 = 70$

Libraries	The total number of books issued	The number of physical books issued	The number of digital books issued
Central	210	70	140
North	280	175	105
South	350	210	140
East	140	35	105
West	210	140	70

Answer: E

The number of physical books issued by the Central library = 70

$\frac{2}{7}$ th of the number of physical books issued = $(\frac{2}{7}) * 70 = 20$

3 times of this value = $3 * 20 = 60$

The total number of books issued by the East library = 140

The required percentage = $(\frac{60}{140}) * 100 = 42.85\%$

Hence, the correct answer is option E.

8. Questions

The total number of books issued = The average number of physical and digital books issued * 2

The number of digital books issued = The total number of books issued - The number of physical books issued

Central

The total number of books issued = $105 * 2 = 210$

The number of physical books issued = 70

The number of digital books issued = $210 - 70 = 140$

North

The total number of books issued = $140 * 2 = 280$

The number of physical books issued = 175

The number of digital books issued = $280 - 175 = 105$

South

The total number of books issued = $175 * 2 = 350$

The number of physical books issued = 210

The number of digital books issued = $350 - 210 = 140$

East

The total number of books issued = $70 * 2 = 140$

The number of physical books issued = 35

The number of digital books issued = $140 - 35 = 105$

West

The total number of books issued = $105 * 2 = 210$

The number of physical books issued = 140

The number of digital books issued = $210 - 140 = 70$

Libraries	The total number of books issued	The number of physical books issued	The number of digital books issued
Central	210	70	140
North	280	175	105
South	350	210	140
East	140	35	105
West	210	140	70

Answer: A

The total number of books issued by the South library = 350

The number of returned physical books = $20\% \text{ of } 350 = (20/100) * 350 = 70$

The number of unreturned digital books = x

According to the question,

$$x + 70 = 95$$

$$x = 95 - 70 = 25$$

Hence, the correct answer is option A.

9. Questions

The total number of books issued = The average number of physical and digital books issued * 2

The number of digital books issued = The total number of books issued - The number of physical books

issued

Central

The total number of books issued = $105 * 2 = 210$

The number of physical books issued = 70

The number of digital books issued = $210 - 70 = 140$

North

The total number of books issued = $140 * 2 = 280$

The number of physical books issued = 175

The number of digital books issued = $280 - 175 = 105$

South

The total number of books issued = $175 * 2 = 350$

The number of physical books issued = 210

The number of digital books issued = $350 - 210 = 140$

East

The total number of books issued = $70 * 2 = 140$

The number of physical books issued = 35

The number of digital books issued = $140 - 35 = 105$

West

The total number of books issued = $105 * 2 = 210$

The number of physical books issued = 140

The number of digital books issued = $210 - 140 = 70$

Libraries	The total number of books issued	The number of physical books issued	The number of digital books issued
Central	210	70	140
North	280	175	105
South	350	210	140
East	140	35	105
West	210	140	70

Answer: C

The number of physical books issued by the East library = 35

The number of digital books issued by the East library = 105

The number of physical books issued by the West library = 140

The number of digital books issued by the West library = 70

Let the late renewal fine for 1 physical book be p and the fine for 1 digital book be d .

$$35p + 105d = 1050 \text{ --- (1)}$$

$$140p + 70d = 2450 \text{ --- (2)}$$

Dividing equation (1) by 35,

$$p + 3d = 30$$

$$p = 30 - 3d$$

Dividing equation (2) by 70:

$$2p + d = 35$$

$$2 * (30 - 3d) + d = 35$$

$$60 - 6d + d = 35$$

$$25 = 5d$$

$$d = 5$$

$$p = 30 - 3 * 5$$

$$p = 30 - 15 = 15$$

The total late renewal fine for 10 physical books = $10 * 15 = 150$

Hence, the correct answer is option C.

10. Questions

The total number of books issued = The average number of physical and digital books issued * 2

The number of digital books issued = The total number of books issued - The number of physical books issued

Central

The total number of books issued = $105 * 2 = 210$

The number of physical books issued = 70

The number of digital books issued = $210 - 70 = 140$

North

The total number of books issued = $140 * 2 = 280$

The number of physical books issued = 175

The number of digital books issued = $280 - 175 = 105$

South

The total number of books issued = $175 * 2 = 350$

The number of physical books issued = 210

The number of digital books issued = $350 - 210 = 140$

East

The total number of books issued = $70 * 2 = 140$

The number of physical books issued = 35

The number of digital books issued = $140 - 35 = 105$

West

The total number of books issued = $105 * 2 = 210$

The number of physical books issued = 140

The number of digital books issued = $210 - 140 = 70$

Libraries	The total number of books issued	The number of physical books issued	The number of digital books issued
Central	210	70	140
North	280	175	105
South	350	210	140
East	140	35	105
West	210	140	70

Answer: B

The number of physical books issued by the North library = 175

40% of the number of physical books = $(40/100) * 175 = 70$

The number of digital books issued by the East library = 105

1/3rd of the number of digital books = $(1/3) * 105 = 35$

The increased value of the digital books = $105 + 35 = 140$

The required fraction = $70/140 = 1/2$

Hence, the correct answer is option B.

11. Questions

The total number of students = 600

The number of students not participating = 80

The number of participating students = $600 - 80 = 520$

The number of students participating only in Football = 100

The number of students participating in both Cricket and Football but not Tennis = $100/2 = 50$

The number of students participating in all three sports = 40

The number of students participating in both Football and Tennis but not Cricket = $40 + 20 = 60$

The number of students participating in both Cricket and Tennis but not Football = 60

Let the number of students participating only in Tennis be t .

The number of students participating only in Cricket = $t + 40$

The Total Participating = $(t + 40) + 100 + t + 50 + 60 + 60 + 40 = 520$

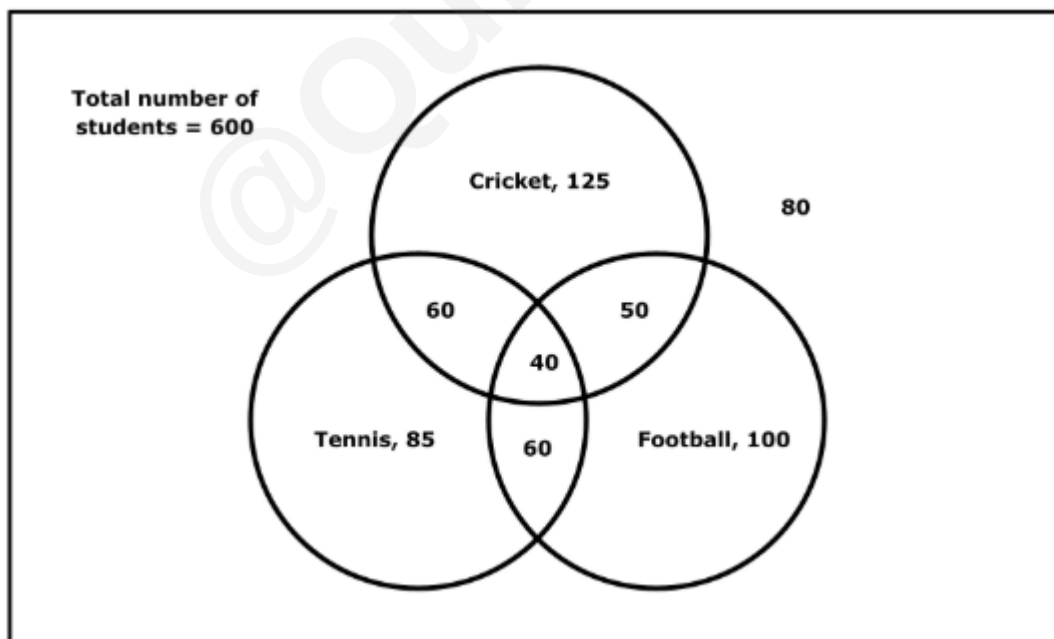
$2t + 350 = 520$

$2t = 170$

$t = 85$

The number of students participating only in Tennis = 85

The number of students participating only in Cricket = $85 + 40 = 125$



Answer: E

The number of students participating only in Cricket = 125

80% of the students participating only in Cricket = $(80/100) * 125 = 100$

The number of students participating in both Cricket and Tennis but not Football = 60

The required ratio = $100:60 = 5:3$

Hence, the correct answer is option E.

12. Questions

The total number of students = 600

The number of students not participating = 80

The number of participating students = $600 - 80 = 520$

The number of students participating only in Football = 100

The number of students participating in both Cricket and Football but not Tennis = $100/2 = 50$

The number of students participating in all three sports = 40

The number of students participating in both Football and Tennis but not Cricket = $40 + 20 = 60$

The number of students participating in both Cricket and Tennis but not Football = 60

Let the number of students participating only in Tennis be t .

The number of students participating only in Cricket = $t + 40$

The Total Participating = $(t + 40) + 100 + t + 50 + 60 + 60 + 40 = 520$

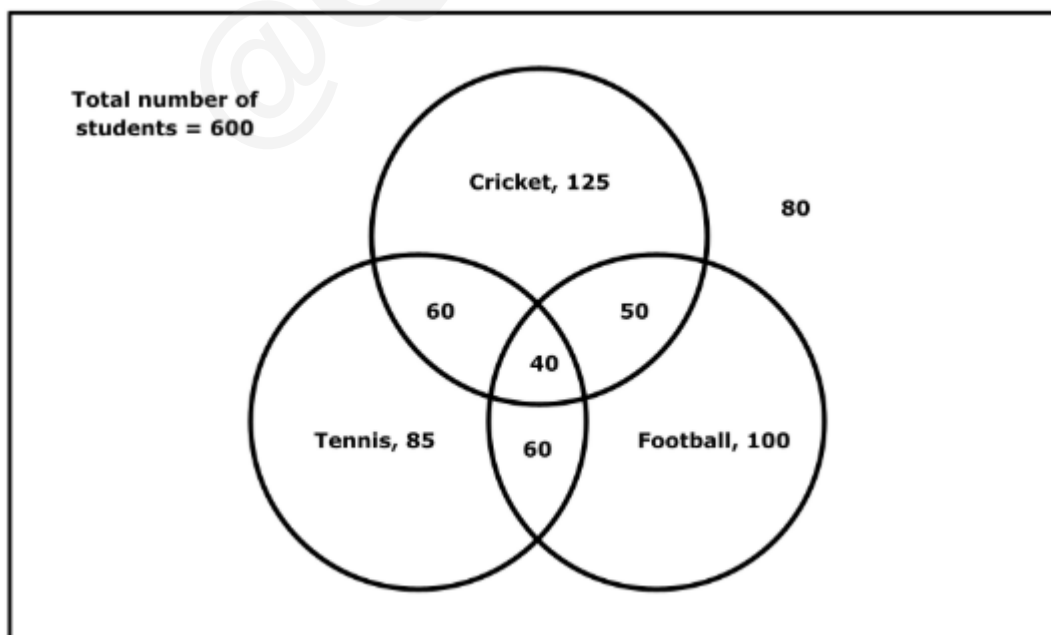
$2t + 350 = 520$

$2t = 170$

$t = 85$

The number of students participating only in Tennis = 85

The number of students participating only in Cricket = $85 + 40 = 125$



Answer: A

The number of students participating only in Tennis = 85

$\frac{4}{5}$ th of the number of students participating only in Tennis = $(\frac{4}{5}) * 85 = 68$

The number of students participating in both Cricket and Football but not Tennis = 50

1.2 times this value = $1.2 * 50 = 60$

The required product = $68 * 60 = 4080$

Hence, the correct answer is option A.

13. Questions

The total number of students = 600

The number of students not participating = 80

The number of participating students = $600 - 80 = 520$

The number of students participating only in Football = 100

The number of students participating in both Cricket and Football but not Tennis = $100/2 = 50$

The number of students participating in all three sports = 40

The number of students participating in both Football and Tennis but not Cricket = $40 + 20 = 60$

The number of students participating in both Cricket and Tennis but not Football = 60

Let the number of students participating only in Tennis be t .

The number of students participating only in Cricket = $t + 40$

The Total Participating = $(t + 40) + 100 + t + 50 + 60 + 60 + 40 = 520$

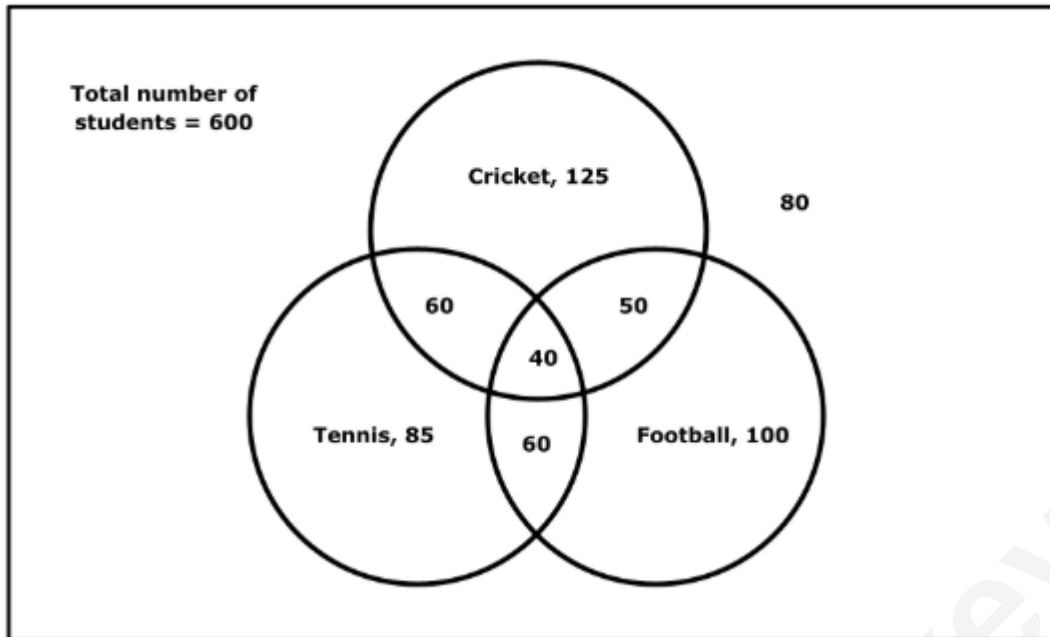
$2t + 350 = 520$

$2t = 170$

$t = 85$

The number of students participating only in Tennis = 85

The number of students participating only in Cricket = $85 + 40 = 125$



Answer: C

The number of students who do not participate in any of these three sports = 80

2.5 times this number = $2.5 \times 80 = 200$

The number of students participating in both Football and Tennis but not Cricket = 60

150% of this number = $(150/100) \times 60 = 90$

The difference = $200 - 90 = 110$

Hence, the correct answer is option C.

14. Questions

The total number of students = 600

The number of students not participating = 80

The number of participating students = $600 - 80 = 520$

The number of students participating only in Football = 100

The number of students participating in both Cricket and Football but not Tennis = $100/2 = 50$

The number of students participating in all three sports = 40

The number of students participating in both Football and Tennis but not Cricket = $40 + 20 = 60$

The number of students participating in both Cricket and Tennis but not Football = 60

Let the number of students participating only in Tennis be t .

The number of students participating only in Cricket = $t + 40$

The Total Participating = $(t + 40) + 100 + t + 50 + 60 + 60 + 40 = 520$

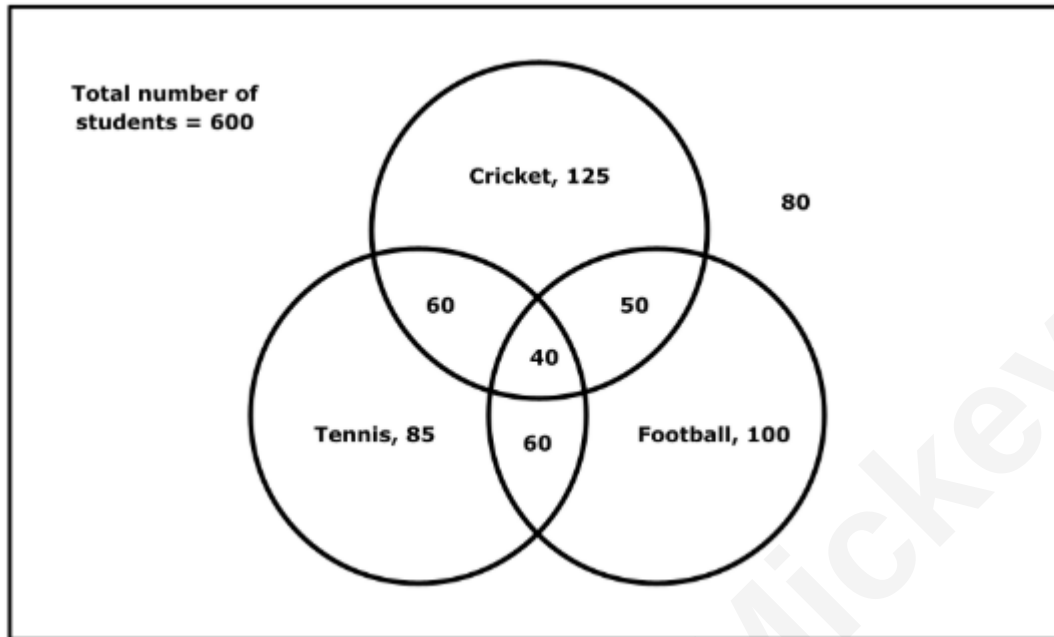
$2t + 350 = 520$

$2t = 170$

$$t = 85$$

The number of students participating only in Tennis = 85

The number of students participating only in Cricket = $85 + 40 = 125$



Answer: B

The number of students participating only in Football = 100

The number of students participating in both Cricket and Tennis but not Football = 60

3.5 times this value = $3.5 * 60 = 210$

According to the given question,

$$y + 100 = 210$$

$$y = 210 - 100 = 110$$

Hence, the correct answer is option B.

15. Questions

Answer: C

The age difference is x , so $A = B + x$.

B's age ' x ' years ago was $B - x$.

We are given: $B - x = A/2$.

Substitute A with $(B + x)$:

$$2 * (B - x) = B + x$$

$$2B - 2x = B + x$$

$$B = 3x.$$

Since $A = B + x$, $A = 4x$.

The age difference (A - B) is x.

C's present age = $2(x) + 8 = 2x + 8$.

Total age = $4x + 3x + 2x + 8 = 62$

$9x + 8 = 62$

$9x = 54$

$x = 6$.

Hence, the correct answer is option C.

16. Questions

Answer: C

Let the selling price of each article be S.

For the first article (20% gain): 120% of CP1 = S, so $CP1 = 5S/6$.

The profit on the first article is $S - 5S/6 = S/6$.

To make no profit or loss overall, the loss on the second article must equal the profit on the first (S/6).

Therefore, the cost price of the second article (CP2) = Selling Price + Loss = $S + S/6 = 7S/6$.

We are given that CP2 is Rs. 1200 more than CP1:

$7S/6 - 5S/6 = 1200$

$2S/6 = 1200$

$S/3 = 1200$

$S = 3600$.

Hence, the correct answer is option C.

17. Questions

Answer: B

Let the total capacity of the vessel be C.

The quantity replaced each time is 20% of C, or C/5.

Replacement formula: Final Milk Quantity = $C * (1 - (C/5)/C)^2 = C * (1 - 1/5)^2$

Final Milk = $C * (16/25) = 0.64C$.

The final quantity of water = Total Capacity - Final Milk Quantity = $C - 0.64C = 0.36C$.

The difference between milk and water quantities is 28 liters,

$0.64C - 0.36C = 28$

$0.28C = 28$

$C = 100$ liters.

The quantity replaced each time is 20% of 100 = 20 liters.

Hence, the correct answer is option B.

18. Questions

Answer: D

P's investment time = 12 months. P's total capital = $p * 12 = 12p$.

Q joins after t months, so his investment time = $(12 - t)$.

R joins after another t months (total $2t$ months late), so his investment time = $(12 - 2t)$.

R's total capital = $3p * (12 - 2t)$.

Since P and R receive equal shares, their total capitals must be equal,

$$12p = 3p * (12 - 2t)$$

$$4 = 12 - 2t$$

$$2t = 8$$

$$t = 4 \text{ months.}$$

Q's investment time = $12 - 4 = 8$ months.

Q's total capital = $2p * 8 = 16p$.

The profit ratio of P:Q:R = $12p:16p:12p = 3:4:3$.

Total ratio parts = 10.

Q's share = $(4/10) * 8000 = \text{Rs. } 3200$.

Hence, the correct answer is option D.

19. Questions

Answer: B

The effective compound interest rate = $A + B + (AB/100)$

The effective compound interest rate for 2 years at 20% = $20 + 20 + (20 * 20/100) = 44\%$.

Interest earned on P = 44% of P = $0.44P$.

The effective simple interest rate for 3 years at 12% = $12 * 3 = 36\%$.

Interest earned on $(P + 2000)$ = 36% of $(P + 2000)$ = $0.36P + 720$.

We are given that the compound interest is Rs. 240 less than the simple interest:

$$(0.36P + 720) - 0.44P = 240$$

$$720 - 0.08P = 240$$

$$0.08P = 480$$

$$P = 480/0.08 = 6000.$$

Therefore, $2P = 2(6000) = 12000$.

Hence, the correct answer is option B.

20. Questions

Answer: C

Let the outer side of the square park be S . Total Area = S^2 .

Since a 2-meter wide path runs inside, the side of the inner square = $S - (2 * 2) = S - 4$.

The area of the inner square = $(S - 4)^2$.

The inner perimeter = $4 * (S - 4) = 4S - 16$.

The area of the path = Total Area - Inner Area = $S^2 - (S - 4)^2$

Path Area = $S^2 - (S^2 - 8S + 16) = 8S - 16$.

According to the given question,

$$8S - 16 = 4 * (4S - 16)$$

$$8S - 16 = 16S - 64$$

$$8S = 48$$

$$S = 6 \text{ m.}$$

The total area of the park = $S^2 = 6^2 = 36 \text{ m}^2$

Hence, the correct answer is option C.

21. Questions

Answer: C

$$(?^2 + 13 * 8) / \sqrt{196} + \sqrt{144} = 27 + 3^3$$

$$(?^2 + 104) / 14 + 12 = 27 + 27$$

$$(?^2 + 104) / 14 + 12 = 54$$

$$(?^2 + 104) / 14 = 54 - 12$$

$$(?^2 + 104) / 14 = 42$$

$$?^2 + 104 = 42 * 14$$

$$?^2 + 104 = 588$$

$$?^2 = 588 - 104$$

$$?^2 = 484$$

$$? = 22$$

Hence, the correct answer is option C.

22. Questions

Answer: B

$$18 * 4 + 20\% \text{ of } 150 + 2(1/4) * ? = 15^2 - 15$$

$$72 + 30 + (9/4) * ? = 225 - 15$$

$$102 + (9/4) * ? = 210$$

$$(9/4) * ? = 210 - 102$$

$$(9/4) * ? = 108$$

$$? = (108 * 4)/9$$

$$? = 12 * 4 = 48$$

Hence, the correct answer is option B.

23. Questions

Answer: D

$$? - 17^2 = (14 * 2 + 12) * (15/2)$$

$$? - 289 = 40 * (15/2)$$

$$? - 289 = 20 * 15$$

$$? - 289 = 300$$

$$? = 300 + 289 = 589$$

Hence, the correct answer is option D.

24. Questions

Answer: A

$$?^3 - 5(1/3) * 27 + 11^2 = 20^2 + 89$$

$$?^3 - (16/3) * 27 + 121 = 400 + 89$$

$$?^3 - (16 * 9) + 121 = 489$$

$$?^3 - 144 + 121 = 489$$

$$?^3 = 512$$

$$? = 8$$

Hence, the correct answer is option A.

25. Questions**Answer: E**

$$480/\sqrt{?} + 11^2 + 8^2 = 14^2 + 29$$

$$480/\sqrt{?} + 121 + 64 = 196 + 29$$

$$480/\sqrt{?} + 185 = 225$$

$$480/\sqrt{?} = 40$$

$$\sqrt{?} = 480/40$$

$$\sqrt{?} = 12$$

$$? = 144$$

Hence, the correct answer is option E.

26. Questions**Answer: B**

$$(?^2 - 22.95 * 4.02)/\sqrt{360.99} = 14.98 + 8.05$$

$$(?^2 - 23 * 4)/19 = 15 + 8$$

$$(?^2 - 92)/19 = 23$$

$$?^2 - 92 = 23 * 19$$

$$?^2 - 92 = 437$$

$$?^2 = 529$$

$$? = 23$$

Hence, the correct answer is option B.

27. Questions**Answer: C**

$$(419.98 - ?)/(11.02/3.99) = 15.95 * 5.03$$

$$(420 - ?)/(11/4) = 16 * 5$$

$$(420 - ?)/(11/4) = 80$$

$$420 - ? = 80 * (11/4)$$

$$420 - ? = 20 * 11$$

$$420 - ? = 220$$

$$? = 200$$

Hence, the correct answer is option C.

28. Questions

Answer: D

$$?^2 + 13.98^2 = 39.95\% \text{ of } 650.05 + 15.97 * 5.02$$

$$?^2 + 14^2 = 40\% \text{ of } 650 + 16 * 5$$

$$?^2 + 196 = 260 + 80$$

$$?^2 = 144$$

$$? = 12$$

Hence, the correct answer is option D.

29. Questions

Answer: A

$$(?^2 + 19)/(25/7) = 16 + 12$$

$$(?^2 + 19)/(25/7) = 28$$

$$?^2 + 19 = 28 * (25/7)$$

$$?^2 + 19 = 4 * 25$$

$$?^2 + 19 = 100$$

$$?^2 = 81$$

$$? = 9$$

Hence, the correct answer is option A.

30. Questions

Answer: E

$$? * 11.95 - \sqrt{1023.98} = 34.98 * 3.99 + 20.02$$

$$? * 12 - \sqrt{1024} = 35 * 4 + 20$$

$$? * 12 - 32 = 140 + 20$$

$$? * 12 = 192$$

$$? = 192/12 = 16$$

Hence, the correct answer is option E.

31. Questions

Answer: B

$$10 + 2^2 = 14$$

$$14 + 3^2 = \mathbf{23}$$

$$23 + 5^2 = 48$$

$$48 + 7^2 = 97$$

$$97 + 11^2 = 218$$

Hence, the correct answer is option B.

32. Questions

Answer: C

$$5 * 2 + 1 = 11$$

$$11 * 3 - 2 = 31$$

$$31 * 4 + 3 = 127$$

$$127 * 5 - 4 = \mathbf{631}$$

$$631 * 6 + 5 = 3791$$

Hence, the correct answer is option C.

33. Questions

Answer: D

50	60	76	98	126
+ 10	+ 16	+ 22	+ 28	
+6	+6	+6		

Hence, the correct answer is option D.

34. Questions

Answer: A

$$48/2 = \mathbf{24}$$

$$24 * 3 = 72$$

$$72/2 = 36$$

$$36 * 3 = 108$$

$$108/2 = 54$$

Hence, the correct answer is option A.

35. Questions

Answer: C

$$100 + (2^3 - 2) = 106$$

$$106 + (3^3 - 3) = 130$$

$$130 + (4^3 - 4) = \mathbf{190}$$

$$190 + (5^3 - 5) = 310$$

$$310 + (6^3 - 6) = 520$$

Hence, the correct answer is option C.

36. Questions

Answer: B

$$15 + 18 = 33$$

$$33 + 21 = 54$$

$$54 + 24 = 78$$

$$78 + 27 = \mathbf{105}$$

$$105 + 30 = 135$$

Hence, the correct answer is option B.

37. Questions

Answer: E

$$100 + (12 * 1) = 100 + 12 = 112$$

$$112 + (12 * 2) = 112 + 24 = 136$$

$$136 + (12 * 3) = 136 + 36 = 172$$

$$172 + (12 * 4) = 172 + 48 = \mathbf{220}$$

$$220 + (12 * 5) = 220 + 60 = 280$$

Hence, the correct answer is option E.

38. Questions

Answer: C

$$8 * 1.5 = \mathbf{12}$$

$$12 * 2 = 24$$

$$24 * 2.5 = 60$$

$$60 * 3 = 180$$

Hence, the correct answer is option C.

39. Questions

Answer: A

$$10 + 1^3 = 10 + 1 = 11$$

$$11 + 3^3 = 11 + 27 = 38$$

$$38 + 5^3 = 38 + 125 = 163$$

$$163 + 7^3 = 163 + 343 = \mathbf{506}$$

Hence, the correct answer is option A.

40. Questions

Answer: D

$$5^2 - 1 = \mathbf{24}$$

$$6^2 - 2 = 34$$

$$7^2 - 3 = 46$$

$$8^2 - 4 = 60$$

$$9^2 - 5 = 76$$

Hence, the correct answer is option D.

41. Questions

Answer: C

I). $3x^2 - 38x + 115 = 0$

$$3x^2 - 15x - 23x + 115 = 0$$

$$3x(x - 5) - 23(x - 5) = 0$$

$$(3x - 23)(x - 5) = 0$$

$$x = 7.66, 5$$

II). $2y^2 - 15y + 28 = 0$

$$2y^2 - 8y - 7y + 28 = 0$$

$$2y(y - 4) - 7(y - 4) = 0$$

$$(2y - 7)(y - 4) = 0$$

$$y = 4, 3.5$$

The comparison:

$$7.66 > 4$$

$$7.66 > 3.5$$

$$5 > 4$$

$$5 > 3.5$$

Therefore, $x > y$.

Hence, the correct answer is option C.

42. Questions

Answer: A

I). $2x^2 + 31x + 119 = 0$

$$2x^2 + 14x + 17x + 119 = 0$$

$$2x(x + 7) + 17(x + 7) = 0$$

$$(2x + 17)(x + 7) = 0$$

$$x = -7, -8.5$$

II). $3y^2 + 25y + 52 = 0$

$$3y^2 + 12y + 13y + 52 = 0$$

$$3y(y + 4) + 13(y + 4) = 0$$

$$(3y + 13)(y + 4) = 0$$

$$y = -4, -4.33$$

The comparison:

$$-7 < -4$$

$$-7 < -4.33$$

$$-8.5 < -4$$

$$-8.5 < -4.33$$

Therefore, $x < y$.

Hence, the correct answer is option A.

43. Questions

Answer: E

I). $2x^2 - 23x + 65 = 0$

$$2x^2 - 13x - 10x + 65 = 0$$

$$x(2x - 13) - 5(2x - 13) = 0$$

$$(x - 5)(2x - 13) = 0$$

$$x = 6.5, 5$$

$$\text{II). } 2y^2 - 19y + 45 = 0$$

$$2y^2 - 10y - 9y + 45 = 0$$

$$2y(y - 5) - 9(y - 5) = 0$$

$$(2y - 9)(y - 5) = 0$$

$$y = 5, 4.5$$

The comparison:

$$6.5 > 5$$

$$6.5 > 4.5$$

$$5 = 5$$

$$5 > 4.5$$

Therefore, $x \geq y$.

Hence, the correct answer is option E.

44. Questions

Answer: C

$$\text{I). } 3x^2 - 7x - 40 = 0$$

$$3x^2 - 15x + 8x - 40 = 0$$

$$3x(x - 5) + 8(x - 5) = 0$$

$$(3x + 8)(x - 5) = 0$$

$$x = 5, -2.66$$

$$\text{II). } y^2 - 14y + 45 = 0$$

$$y^2 - 9y - 5y + 45 = 0$$

$$y(y - 9) - 5(y - 9) = 0$$

$$(y - 5)(y - 9) = 0$$

$$y = 9, 5$$

The comparison:

$$5 < 9$$

$$5 = 5$$

$$-2.66 < 9$$

$$-2.66 < 5$$

Therefore, $x \leq y$.

Hence, the correct answer is option C.

45. Questions

Answer: B

I). $2x^2 - 27x + 88 = 0$

$$2x^2 - 16x - 11x + 88 = 0$$

$$2x(x - 8) - 11(x - 8) = 0$$

$$(2x - 11)(x - 8) = 0$$

$$x = 8, 5.5$$

II). $y^2 - 13y + 42 = 0$

$$y^2 - 7y - 6y + 42 = 0$$

$$y(y - 7) - 6(y - 7) = 0$$

$$(y - 6)(y - 7) = 0$$

$$y = 7, 6$$

The comparison:

$$8 > 7$$

$$8 > 6$$

$$5.5 < 7$$

$$5.5 < 6$$

Therefore, $x = y$ or the relationship cannot be established.

Hence, the correct answer is option B.